Miru Hong

PHD APPLICANT · FOOTBALL AI · MACHINE LEARNING

Seoul, South Korea
mirunoyume@uos.ac.kr · favoruu39@gmail.com · hmiru.github.io
Google Scholar (https://scholar.google.com/citations?user=jsxOv_kAAAAJ)
LinkedIn (https://www.linkedin.com/in/miru-hong-64b737285/)

01 · RESEARCH INTERESTS

I build player-conditioned football world models that generate on-ball events and off-ball trajectories in one framework. I want player representations that hold steady when the team, the manager and the match state around a player change.

Football World Models, Joint Event and Trajectory Generation, Disentangled Player Representations, Player-Conditioned Simulation, Tactical Fit and Transfer Evaluation, Sports Analytics

02 · EDUCATION

University of Seoul, Dept. of Artificial Intelligence — 2025–2027 (expected)

M.S. in Artificial Intelligence · Seoul, South Korea

- Advisor: Prof. Sang-Ki Ko
- Thesis (in progress): generative simulation of football from event and tracking data

Inha University, Dept. of Computer Science and Engineering — 2019–2025

B.S. in Computer Science and Engineering · Incheon, South Korea

03 · PUBLICATIONS

∗ equal contribution † corresponding author

PEER-REVIEWED

01 **Miru Hong**, Minho Lee, Geonhee Jo, Hyeokje Cho, Hyunsung Kim, Pascal Bauer, Sang-Ki Ko†. "Modeling Matches as Language: A Generative Transformer Approach for Counterfactual Player Valuation in Football." *ECML-PKDD*, 2026. Applied Data Science Track. [paper] (https://arxiv.org/abs/2603.15212)

02 Minho Lee, Geonhee Jo, **Miru Hong**, Pascal Bauer, Sang-Ki Ko†. "Valuing La Pausa: Quantifying Optimal Pass Timing Beyond Speed." *MIT Sloan Sports Analytics Conference*, 2026. [paper] (https://www.sloansportsconference.com/research-papers/valuing-la-pausa-quantifying-optimal-pass-timing-beyond-speed) Research Paper Competition Finalist

03 **Miru Hong**∗, Minho Lee∗, Geonhee Jo, Jae-Hee So, Pascal Bauer, Sang-Ki Ko†. "ScoutGPT: Capturing Player Impact from Team Action Sequences Using GPT-Based Framework." *Hudl Conference*, 2025. [paper] (https://arxiv.org/abs/2512.17266)

04 Geonhee Jo, **Miru Hong**, Han-Jun Choi, Minho Lee, Pascal Bauer, Sang-Ki Ko†. "Imputing Multi-Agent Trajectories from Event and Snapshot Data in Soccer." *CIKM*, 2025. [paper] (https://doi.org/10.1145/3746252.3760868)

05 Minho Lee, Geonhee Jo, **Miru Hong**, Pascal Bauer, Sang-Ki Ko†. "exPress: Contextual Valuation of Individual Players within Pressing Situations in Soccer." *MIT Sloan Sports Analytics Conference*, 2025. [paper] (https://www.sloansportsconference.com/research-papers/contextual-valuation-of-individual-players-within-pressing-situations-in-football)

06 Minho Lee, Geonhee Jo, **Miru Hong**, Pascal Bauer, Sang-Ki Ko†. "Contextual Evaluation of Individual Contributions from Pressing Situations in Football." *MLSA Workshop @ ECML-PKDD*, 2025. [paper] (https://link.springer.com/chapter/10.1007/978-3-032-15165-0_3)

IN PREPARATION

01 **Miru Hong**, et al. Generative simulation of football from event and tracking data. In preparation, 2026. Extends the event-only model with tracking data for trajectory generation, higher-fidelity simulation, and player embeddings that transfer across clubs. *Also the basis of the M.S. thesis.*

DOMESTIC CONFERENCES (KOREAN)

01 **Miru Hong**, Minho Lee, Sang-Ki Ko†. "Soccer Cross Outcome Prediction with 3D Ball Trajectories and Contextual Information." *Korea Computer Congress (KCC)*, 2026. `Outstanding Paper Award`

02 Geonhee Jo, **Miru Hong**, Sang-Ki Ko†. "Player Position Prediction Using Soccer Event and Snapshot Data." *Korea Computer Congress (KCC)*, 2025.

---

04 · RESEARCH EXPERIENCE

CIDA Lab, University of Seoul · Graduate Research Assistant

2025–present

Computational Intelligence and Data Analytics Laboratory · Advisor: Prof. Sang-Ki Ko

- Built *ScoutGPT*, an autoregressive Transformer that treats a football match as language. The model splits each event into discrete attributes: actor, location, and action type. It predicts the next attribute from the previous ones, the lineup, and the match context.
- Trained the model on 6.1M K League events from 2,283 matches between 2021 and 2025. Added a value-prediction head aligned to VAEP. The model performs better than CatBoost, LSTM, and sequence-model baselines on next-event prediction and goal prediction.
- Added counterfactual transfer simulation with Monte Carlo sampling. The method puts a player into a new lineup and measures the change in offensive progression and goal probability. It reduces post-transfer VAEP forecast error against a naive carry-over baseline.
- Current work extends the framework to tracking data for trajectory generation and player embeddings that transfer across clubs. Role-masking makes sure that positional role does not become a shortcut in cross-season player retrieval.

Pixelscope · Industry Collaboration

2026, 3 months

- Built a pipeline that derives structured event data from raw player tracking data, reaching F1 > 0.90 on most kick events and F1 > 0.85 on ball control.

K League · Data Analysis Contributor

2025–2026

- Contributed to the 2025 K League end-of-season analytical review (Technical Study Group report).
- Taught at the K League Data Academy in 2025 and 2026, and produced its teaching material.

---

05 · TEACHING EXPERIENCE

Football Data Analytics · Teaching Assistant

Fall 2025 & 2026

University of Seoul · general elective

---

06 · AWARDS & HONORS

Research Paper Competition Finalist, MIT Sloan Sports Analytics Conference

2026

for "Valuing La Pausa: Quantifying Optimal Pass Timing Beyond Speed"

Outstanding Paper Award, Korea Computer Congress (KCC)

2026

Korean Institute of Information Scientists and Engineers (KIISE)

---

07 · TECHNICAL SKILLS

Programming	Python (primary), C++
ML / Data	PyTorch, XGBoost, CatBoost, scikit-learn, pandas, NumPy

Methods	Autoregressive sequence models, Transformers, representation and disentangled learning, causal inference, spatiotemporal modeling
Football data	Event data (VERSA, VAEP), tracking data, event–tracking synchronisation, K League and Bundesliga datasets
Tools	Git, Docker (multi-GPU lab cluster), Linux, LaTeX
Languages	Korean (native), English (IELTS 7.0), Spanish (DELE A2)

08 · TALKS & PRESENTATIONS

"Modeling Matches as Language: A Generative Transformer Approach for Counterfactual Player Valuation in Football" 2026

ECML-PKDD, Applied Data Science Track · oral and poster

"Modeling Matches as Language: A Generative Transformer Approach for Counterfactual Player Valuation in Football" 2026

NU Sports ML Workshop, Nagoya University (organiser: Prof. Keisuke Fujii) · preview of the ECML-PKDD paper

"Imputing Player Coordinates and Velocities using Event Data in Football" 2025

NU Sports ML Workshop, Nagoya University (organiser: Prof. Keisuke Fujii)

09 · REFERENCES

Available upon request.